



Department of Computer Science and Engineering
Green University of Bangladesh
Midterm Examination, Fall 2024
CSE 303, Microprocessors and Microcontrollers



Duration: 1.5 Hours

Total Marks: 30

Answer all of the following questions.

(The CO# along with each sub-question represents the mapping of the questions onto the expected outcomes of the course provided in the course outline.)

Question 1.

a.

CO1 1+3 Marks

Explain the purpose of the stack in a microprocessor. Describe the steps involved in the PUSH PSW operation, along with an appropriate diagram.

— OR —

Critically compare the internal architecture features of the 8085 and 8086 microprocessors. Highlight the distinctive characteristics that set these two microprocessors apart in terms of their internal structures.

b.

CO1 3 Marks

Predict the ending address of 8KB memory with the starting address 4A00 H.

c.

CO1 3 Marks

Draw the timing and state diagram for the opcode fetch cycle in the operation of the 8085 microprocessor.

Question 2.

a.

CO2 3 Marks

Suppose a 2K*8 ROM is connected with an 8085 microprocessor from an address range of 5000H to 57FFH. Detect the chip select logic for the given ROM.

— OR —

Detect the number of 256x8 ROM chips required to design 16KB of memory.

b.

CO2 2 Marks

Consider the below states of the microprocessor pins. Analyze what this configuration indicates:

$$\begin{aligned}\overline{IO/M} + \overline{RD} &= 1 + 0 \\ \overline{IO/M} + \overline{WR} &= 0 + 1\end{aligned}$$

CO2 5 Marks

Determine the count value, N, for the following code to produce a time delay of 0.6 milliseconds in a microprocessor with an 8 MHz quartz crystal clock source.

```
CALL addr16
MVI C,0
MVI D,N
LOOP: DCR D
ICR C
JNZ LOOP
RET
```

where, the T states required for each instruction are as follows-

Instructions	T-State required for execution of an instruction
CALL addr16	18
MVI C,0	7
MVI D,N	7
DCR D	4
ICR C	4
JNZ LOOP	10 (or 7)
RET	10

Question 3.

a.

CO1 3 Marks

Show the status of the flag register after executing the following instruction line by line-

```
MVI A, 2E H
MVI B, F3 H
ADD B
HLT
```

b.

CO1 3 Marks

Given the binary data 10011100 with the initial Carry Flag (CY) set to 1, perform the following operations:

- Apply the RAR instruction twice.
- Apply the RLC instruction once on the result obtained after the second RAR operation

After each operation, show the resulting data and the status of the Carry Flag (CY).

c.

CO1 4 Marks

Discuss the different addressing modes of 8085 microprocessors with examples.

— OR —

Differentiate between maskable and non-maskable interrupts in 8085 microprocessors and explain how the 8085 microprocessor handles multiple interrupts occurring simultaneously.

□ End of Question □